

p(Yes)

new instance x outlook = Rainy, temperature = Cool
humidity = Normal, wind = True.

$$(1) p(\text{outlook} = \text{Rainy} / \text{play} = \text{yes}) = \frac{p(\text{outlook} \cap \text{yes})}{p(\text{yes})}$$

$$= \frac{3}{14} \times \frac{14}{9} = \frac{3}{9}$$

$$(2) p(\text{Temperature} = \text{Cool} / \text{play} = \text{yes}) = \frac{p(\text{Cool} \cap \text{yes})}{p(\text{yes})} = \frac{\frac{3}{14}}{\frac{9}{14}} = \frac{3}{9}$$

$$(3) p(\text{humidity} = \text{Normal} / \text{play} = \text{yes}) = \frac{p(\text{Normal} \cap \text{yes})}{p(\text{yes})} = \frac{\frac{6}{14}}{\frac{9}{14}} = \frac{6}{9}$$

$$(4) p(\text{wind} = \text{True} / \text{play} = \text{yes}) = \frac{p(\text{True} \cap \text{yes})}{p(\text{yes})} = \frac{\frac{3}{14} \times \frac{14}{9}}{\frac{9}{14}} = \frac{3}{9}$$

$$(5) p(\text{play} = \text{yes}) = \frac{9}{14}$$

$$P(\text{No})$$

$$(1) P(\text{outlook} = \text{rainy} / \text{No}) = \frac{2}{14} \times \frac{14}{5} = \frac{2}{5}$$

$$(2) P(\text{Temp} = \text{cool} / \text{No}) = \frac{P(\text{cool} \cap \text{No})}{P(\text{No})}$$

$$= \frac{1}{14} \times \frac{14}{5} = \frac{1}{5}$$

$$(3) P(\text{Humidity} = \text{Normal} / \text{No}) = \frac{P(\text{Normal} \cap \text{No})}{P(\text{No})}$$

$$= \frac{1}{14} \times \frac{14}{5} = \frac{1}{5}$$

$$(4) P(\text{wind} = \text{True} / \text{No}) = \frac{P(\text{True} \cap \text{No})}{P(\text{No})}$$

$$= \frac{\frac{3}{14}}{\frac{15}{14}} = \frac{3}{5}$$

$$(5) P(\text{play} = \text{No}) = \frac{5}{14}$$

$$p(x|\text{play} = \text{yes}) \cdot p(\text{play} = \text{yes}) = \frac{2}{9} \times \frac{2}{9} \times \frac{2}{9} \times \frac{2}{9} \times \frac{1}{14} = \frac{1}{63} = 0.00158$$

$$p(x|\text{play} = \text{no}) \cdot p(\text{play} = \text{No}) = \frac{2}{5} \times \frac{1}{5} \times \frac{1}{5} \times \frac{3}{5} \times \frac{1}{14} = \frac{9}{875} = 0.00342$$

$$p(x) = 0.00158 + 0.00342 = 0.005$$

$$p(\text{play} = \text{yes}|x) = \frac{p(x|\text{play} = \text{yes}) \cdot p(\text{play} = \text{yes})}{p(x)} = \frac{0.00158}{0.005} = 0.316$$

$$p(\text{play} = \text{No}|x) = \frac{p(x|\text{play} = \text{no}) \cdot p(\text{play} = \text{No})}{p(x)} = \frac{0.00342}{0.005} = 0.684$$

with probability 0.684 0.68.4% - golf will be played is "No".